SOFIA VALDEZ

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EDUCATION

The University of Texas at Austin – Austin, Texas
Bachelor of Science, Mechanical Engineering **May 2020**

- Minor: McCombs Business Foundations
- GPA: 3.96

PhD, Mechanical Engineering – Manufacturing, Design & AI **August 2025**

- GPA: 3.80

The University of Texas at Austin – Freiburg, Germany
Study Abroad **May 2017 - June 2017**

- Completed 3 credit hours of *Introduction to Engineering Design & Graphics*

EXPERIENCE

The University of Texas at Austin – Austin, Texas **August 2020 – August 2025**
Graduate Researcher

- Designed and developed an AI-powered interactive design interface for exploring and editing topology optimized structures in real time using deep learning latent spaces.
- Collaborated with a furniture retailer and a 3D printing manufacturer to translate the research into a real-world application, enabling customers to design and 3D print custom furniture using recycled plastic.
- Conducted 130+ user interviews and studies with customers, students, and industry partners to understand human-AI interaction and inform iterative improvements to AI models, tools, and interface design.

Moxee Technologies (Panda Mobile Brand) – Bellevue, Washington **September 2024 - March 2025**
UX Research & Product Design Intern (Part-Time)
Project 1: Onboarding-to-Checkout Conversion Optimization (Assigned Project & Independently Led)

- Conducted 7 usability interviews with primary account holders and synthesized findings into a customer needs framework, which guided onboarding funnel redesign and competitive benchmarking.
- Designed a new end-to-end onboarding flow in Figma, resulting in a 2x increase in checkout conversion.

Project 2: Customer Insights for Product & Marketing Strategy (Self-initiated Project & Independently Led)

- Recruited 50+ Chinese international students in one day using incentive-based system and completed 20+ hour-long interviews to uncover high-value customer insights.
- Developed actionable recommendations based on customer research, many of which were implemented, guiding marketing strategy (campaigns, events, and referral incentives), product decisions (pricing and feature prioritization), and brand and competitive positioning (differentiation and messaging).

The University of Texas at Austin – Austin, Texas **August 2021 - May 2022**
Design Methodology Teaching Assistant

- Mentored student teams on company-sponsored design projects, guiding them through the full design process from customer and background research to ideation, prototyping, and functional design.

National Renewable Energy Laboratory – Austin, Texas **June 2021 - August 2021**
Data Science Intern

- Built a machine learning model to predict power usage effectiveness values months in advance, enabling more efficient energy planning and reduced energy waste.
- Reduced latency of a data center energy balance algorithm from hours to seconds by enabling real-time ingestion of power consumption data and visualization of energy flow across devices.

Halliburton – Elmendorf, Texas **June 2019 - August 2019**

Product Design Intern

- Designed and implemented cost-saving engineering solutions for frac manifold trailer cleanup, with potential annual savings at Elmendorf facility of $200k.
- Conducted 50+ interviews across field operators, technicians, engineers, and real estate managers to identify workflow inefficiencies and co-design solutions.

ACTIVITIES

Academic Design Projects – Austin, Texas **August 2019 - May 2022**

- Completed project-based design coursework in *Design Methodology I & II, Reverse Engineering Design, Medical Device Design*, applying the full user-centered design process from background research (patent searches, competitor analysis) and customer needs analysis (150+ interviews) to structured ideation (House of Quality, 6-3-5 brainwriting, TRIZ, morph matrices, Pugh charts), prototyping, testing, and building functional products. Projects included a 3D food printer, hair-dryer brush, menstrual pain-relief device, and a wearable sunshade device.

Personal Design Projects – Austin, Texas **August 2020 - May 2023**

- Independently pursued entrepreneurial and product design projects, including an anti–package-theft device pitched in a Kendra Scott competition, a first-of-its-kind skincare tool combining ice therapy and gua sha techniques, an Amazon best-selling manifestation journal, and an Etsy product line that received multiple "Etsy's Pick" distinctions.

ADDITIONAL

- **Honors & Fellowships:** National Science Foundation Graduate Research Fellow; GEM Fellow; Virginia and Ernest Cockrell Jr. Fellow; Hispanic Scholarship Fund Scholar (2020–2025); Valedictorian, John B. Alexander High School, Class of 2016 (Ranked #1 of ~800)
- **Leadership:** Graduate Liaison of Society of Hispanic Professional Engineers; Mentor of Freshman Introduction to Research in Engineering; Tutor for Calculus I and II
- **Language Fluency**: Native English and Spanish
- **Technical Skills:** Python, R, SQL, MATLAB, data analysis, data visualization, machine learning, deep learning
- **Design & Research Skills:** Figma, interaction design, wireframing, prototyping, user flows, information architecture, usability testing, qualitative user research, contextual inquiry, customer interviews, user journey mapping, affinity mapping and thematic analysis, research synthesis, customer needs synthesis, competitive analysis
- **Publications:**
  - S. Valdez, C. Seepersad and S. Kambampati, "A Framework for Interactive Structural Design Exploration," in *IDETC/CIE Design Automation Conference*, Virtual Conference, 2021.
  - S. Valdez, C. Seepersad and N. Rodriguez, "Latent Variable Representations for Interactive Structural Design Exploration," in *IDETC/CIE Design Automation Conference*, 2022.
- **Work Eligibility:** Eligible to work in the U.S. with no restrictions